

Awakening Cortical Areas via Human-LLM Co-Design: A Paradigm for Embodied Intelligence Beyond Local Learning

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Abstract

We introduce a paradigm for building intelligent embodied agents that does not rely on local learning or gradient-based optimization, but on the progressive awakening of dormant cortical areas through human-LLM co-design. Working within a biologically grounded substrate of cortical microcircuits organized into cortical areas, we show that a single unassigned area—N+4—when equipped with threat hysteresis, resolves the “ghost predator” problem and drives a qualitative transition in long-horizon behavioral success, operationalized as nest completion, from 0% to 100%. The mechanism of N+4 was not learned through synaptic adaptation: it was identified through a constrained dialogue between human experts and a large language model under biological and architectural constraints on microcircuit organization. The result suggests that scalable autonomy may not always require enlarging existing representational modules or learning additional parameters, but may instead emerge through the functional allocation of computational resources that are already available within an architecture. We formulate the *Progressive Awakening Hypothesis*: dormant areas (N+5, N+6, ...) may unlock computational capacities inaccessible to the preceding architecture. This reframes one route to intelligence as a process of architectural awakening rather than exclusively local learning, and offers a testable design principle for embodied agents operating in hostile environments.

1 Introduction

From local learning to functional allocation

Most contemporary artificial-intelligence systems acquire new capabilities by modifying parameters through learning or optimization. In biological and brain-inspired systems, however, an additional possibility exists: a computational resource may already be structurally present but functionally unassigned, with behavior changing when that resource becomes recruited into the active architecture.

The present work investigates this second possibility. Rather than asking how an agent can learn a new policy by modifying synaptic weights, we ask a different question:

What computation should an already available cortical area perform when the existing architecture encounters a behavioral failure?

We study this question in an embodied agent controlled by populations of cortical microcircuits (CMs) organized into functionally specialized cortical areas (CAs). The general theoretical motivation derives from the modular organization of neocortical circuitry proposed by Mountcastle (Mountcastle, 1997) and subsequently formalized through canonical cortical microcircuit descriptions (Douglas & Martin, 2004). Computational studies have shown that rich dynamics can emerge from interactions among local microcircuit subsystems (Haeusler & Maass, 2007), while canonical microcircuit motifs have also been incorporated into hierarchical predictive-processing models (Bastos et al., 2012). Biophysically detailed work has further demonstrated the computational richness of cell-type-specific cortical circuitry (Potjans & Diesmann, 2014; Markram et al., 2015).

Our model is not a biophysical reconstruction of cortex. It is a simplified, biologically inspired computational architecture in which repeated CMs form directionally tuned cortical areas and compete continuously for behavioral influence. This architecture builds on earlier work using cooperative neural agents for embodied control (Chang et al., 2010) and on our previous work combining cortical microcircuits with memory-oriented computational structures (Chang et al., 2025).

A defining property of the present experiment is that the relevant synaptic-like weights remain fixed during a simulation run. No backpropagation, reinforcement-learning update, gradient descent, or online synaptic-plasticity rule is used to acquire the N+4 function. The experimental intervention is therefore not parameter learning. It is *functional allocation*.

The long-horizon problem

An embodied agent operating in a hostile environment faces competing temporal demands. Immediate threats require rapid reaction, whereas tasks such as collecting material and completing a nest require behavioral organization across much longer intervals. The

presence of the required behavioral machinery does not guarantee that the machinery can be successfully exploited.

In the baseline architecture, the agent can orient toward resources, avoid boundaries, seek home, respond to a predator, and execute a repeated material–nest construction sequence. Nevertheless, intermittent predation repeatedly interrupts this longer-horizon behavior.

The important failure occurs when the predator leaves the effective threat-perception region. A purely reactive controller rapidly loses the state that justified defensive behavior. The absence of current perceptual evidence is therefore treated as if it implied safety. When the predator re-enters the relevant interaction range, the creature can be caught before its long-horizon construction behavior succeeds.

We refer to this temporal failure as the *ghost predator problem*.

The distinction is subtle but important. The missing information is not necessarily a richer representation of *what* the predator is. The architecture already reacts to the predator while it is perceptually relevant. The missing variable is *how long the recent threat should continue to influence behavior*.

This temporal framing is consistent with accounts of working memory in which information is sustained through persistent neural activity rather than through a static symbolic store (Curtis & Sprague, 2021), and with computational accounts that connect persistent internal states to planning across time (Ehrlich & Murray, 2022).

Human–LLM co-design as a functional search process

The available architecture contained an additional cortical area, N+4, with no assigned behavioral role. Rather than allowing unrestricted redesign, the human authors constrained the problem:

- no additional sensor could be introduced;
- the environmental rules could not be changed;
- the existing functional areas could not be replaced;
- no external symbolic memory could be added;
- the available N+4 cortical resource had to be used.

Under these constraints, a human–LLM dialogue was used to search for a functionally sufficient computation for N+4. The critical proposal was temporal persistence of threat through an asymmetric dynamical state: fast activation when danger is perceived and slower decay after immediate evidence decreases.

The contribution of this study is therefore threefold.

First, we experimentally evaluate threat hysteresis as a minimal temporal mechanism for preserving the behavioral relevance of a recent threat. Second, we use this case to demonstrate constrained human–LLM co-design as a method for assigning function to an otherwise unassigned cortical resource. Third, we formulate a broader hypothesis: progressively available cortical areas may enable qualitatively new computation when each is assigned a function that resolves a failure exposed by the preceding architecture.

We call this the *Progressive Awakening Hypothesis*.

2 Methods

World and agent

The agent inhabits a two-dimensional square environment

$$[-1, 1] \times [-1, 1].$$

The environment contains four behaviorally relevant entities: food, home, a predator, and construction material. Food satisfies the hunger drive. Home provides a safety region. The predator pursues the creature while it is away from home. Construction material can be collected, carried, and deposited at a nest location.

The creature is represented by a position

$$\mathbf{p}(t) \in \mathbb{R}^2$$

and an orientation $\theta(t)$.

Two principal homeostatic variables are maintained: hunger $h \in [0, 1]$ and safety deficit $s \in [0, 1]$. In the implemented simulation, hunger increases by 0.0015 per frame,

$$h(t+1) = \text{clip}(h(t) + 0.0015, 0, 1), \quad (1)$$

whereas safety deficit increases by 0.0020 per frame while the creature is outside the home region and returns to zero when the creature occupies the home region.

The home radius is

$$r_{\text{home}} = 0.25.$$

The construction task requires repeated interaction with material and the nest. The material load is represented continuously by

$$L(t) \in [0, 1],$$

rather than by a binary possession flag.

All CM weights are initialized from fixed parameter ranges and remain unchanged during a run. Consequently, behavioral changes reported below do not result from within-run adjustment of synaptic weights.

Cortical architecture

The complete controller contains four functional areas. The first three constitute the baseline architecture; N+4 is the additional area studied in the present work.

Each cortical area contains

$$N = 25$$

directionally tuned CMs, with preferred angles

$$\theta_i = \frac{2\pi i}{N}, \quad i = 0, \dots, N - 1. \quad (2)$$

The activity vector of cortical area a is denoted

$$\mathbf{a}^a \in \mathbb{R}^N.$$

Competition among CMs is implemented through continuous softmax relaxation,

$$\mathbf{a}^a(t+1) = (1 - \alpha_a)\mathbf{a}^a(t) + \alpha_a \text{softmax}\left(\frac{\mathbf{E}^a(t)}{T_a}\right), \quad (3)$$

where $\mathbf{E}^a$ is the energy vector of area a , α_a controls relaxation speed, and T_a controls the sharpness of competition.

This softmax rule is an engineering approximation to graded competitive dynamics rather than a claim that biological cortex literally implements a softmax operation. Lateral-inhibition and winner-take-all motifs provide the biological and computational motivation for such competition ([Amari, 1977](#); [Coultrip et al., 1992](#)).

The parameters used in the four areas are summarized in Table 1.

Table 1: Competitive-dynamics parameters used for each cortical area.

Area	Function	α_a	T_a
N+1	Motor / homeostatic behavior	0.35	0.25
N+2	Navigation / border avoidance	0.40	0.15
N+3	Sensorimotor / nest construction	0.35	0.20
N+4	Threat hysteresis / escape	0.35	0.20

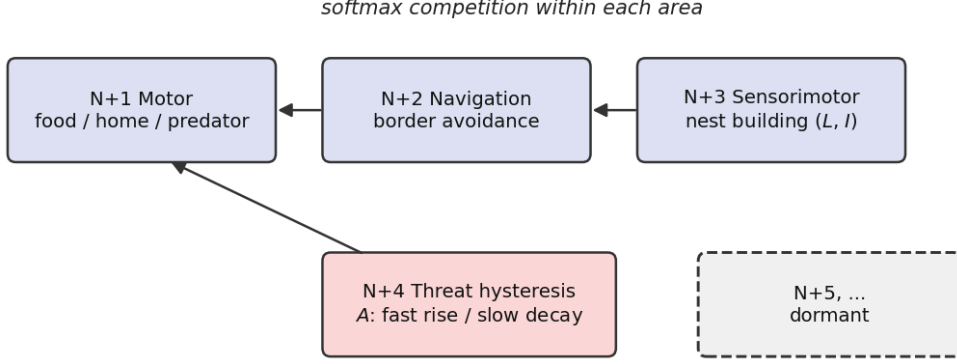


Figure 1: Schematic of the cortical architecture. Areas N+1, N+2, and N+3 form the baseline controller. N+4 is an available cortical area assigned a threat-hysteresis function through the human-LLM co-design process. The dashed region represents subsequently available areas considered by the Progressive Awakening Hypothesis.

Area N+1: Motor and homeostatic control

Area N+1 integrates hunger, safety demand, and immediately relevant predator information into a primary directional motor command.

For CM i ,

$$\begin{aligned}
 E_i^{N+1} = & h_{\text{eff}} w_i^{\text{food}} \cos(\phi_{\text{food}} - \theta_i) \\
 & + s_{\text{eff}} w_i^{\text{home}} \cos(\phi_{\text{home}} - \theta_i) \\
 & - d_{\text{eff}} w_i^{\text{pred}} \cos(\phi_{\text{pred}} - \theta_i) + \epsilon_i,
 \end{aligned} \tag{4}$$

where ϕ_x denotes the bearing toward entity x , d_{eff} denotes the immediately effective threat signal, and ϵ_i represents continuous exploratory noise.

This area therefore provides reactive behavior while the threat is currently relevant.

Area N+2: Navigation

Area N+2 contributes boundary avoidance. For border stress $b \in [0, 1]$,

$$E_i^{N+2} = -b w_i^{\text{border}} \cos(\phi_{\text{border}} - \theta_i) + \epsilon_i. \tag{5}$$

This area prevents the primary motor command from driving the creature persistently into environmental boundaries.

Area N+3: Sensorimotor construction

Area N+3 organizes the repeated material-nest sequence required for construction.

144 Its activity depends on material availability m , continuous material load L , and
 145 developmental construction drive I :

$$E_i^{N+3} = (1 - L)mw_i^{\text{mat}} \cos(\phi_{\text{mat}} - \theta_i)(0.8 + 0.5I) \\ + Lw_i^{\text{nest}} \cos(\phi_{\text{nest}} - \theta_i)(0.7 + 0.5I) + \epsilon_i. \quad (6)$$

146 When the creature is not carrying material, N+3 biases behavior toward available
 147 material. As the continuous load increases, the same area progressively biases behavior
 148 toward the nest.

149 This produces a temporally extended behavioral sequence without modifying CM
 150 weights during execution.

151 **Area N+4: threat hysteresis**

152 The functional role assigned to N+4 is temporal persistence of recent threat.

153 Let

$$d_{\text{pred}}(t) = \|\mathbf{p}(t) - \mathbf{p}_{\text{pred}}(t)\|.$$

154 Current threat relevance is represented by the continuous perception signal

$$\pi(t) = \sigma \left(\frac{r_{\text{percept}} - d_{\text{pred}}(t)}{\varepsilon} \right), \quad (7)$$

155 where

$$\sigma(x) = \frac{1}{1 + e^{-x}},$$

$$r_{\text{percept}} = 0.45,$$

156 and

$$\varepsilon = 0.04.$$

157 The latter corresponds to the slope used by the implemented sigmoid threat gate.

158 The N+4 alert state

$$A(t) \in [0, 1]$$

159 evolves according to

$$\frac{dA}{dt} = k_{\text{rise}}\pi(t)(1-A) - \frac{A}{\tau_{N4}}(1-\pi(t)), \quad \tau_{N4} > 0, \quad (8)$$

with

$$k_{\text{rise}} = 0.9.$$

When threat is strongly perceived,

$$\pi(t) \rightarrow 1,$$

the first term rapidly increases A . When current threat evidence decreases,

$$\pi(t) \rightarrow 0,$$

the second term produces approximately exponential decay with time constant τ_{N4} .

Thus,

fast activation + slow decay = threat hysteresis.

The principal condition studied below uses

$$\tau_{N4} = 16$$

frames.

The directional energy landscape of N+4 is

$$E_i^{N+4} = A w_i^{\text{escape}} \cos(\phi_{\text{escape}} - \theta_i) + \epsilon_i, \quad (9)$$

where

$$\phi_{\text{escape}} = \phi_{\text{pred}} + \pi \quad (10)$$

is the direction opposite the predator bearing.

Figure 2 illustrates the temporal mechanism.

Population decoding and motor integration

Each cortical area produces a two-dimensional population vector

$$\mathbf{m}^a = \sum_{i=1}^N a_i^a \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix}. \quad (11)$$

The N+4 contribution to motor behavior is continuously weighted by its persistent

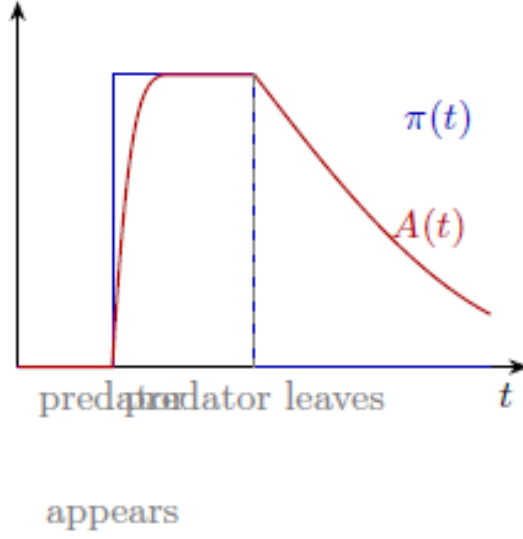


Figure 2: Threat hysteresis. The instantaneous threat signal $\pi(t)$ rises while the predator lies within the effective perception region. The alert state $A(t)$ rises rapidly but decays more slowly after immediate threat evidence decreases. N+4 therefore allows a recent threat to continue influencing behavior after the instantaneous threat signal has weakened.

174 alert state:

$$\lambda_4(t) = \text{clip}(0.65A(t), 0, 0.65). \quad (12)$$

175 Let $\lambda_3(t)$ denote the pre-existing weighting assigned to the construction area N+3.

176 The primary motor signal is then formed as

$$\mathbf{m}_{\text{primary}} = (1 - \lambda_3 - \lambda_4) \mathbf{m}^{N+1} + \lambda_3 \mathbf{m}^{N+3} + \lambda_4 \mathbf{m}^{N+4}, \quad (13)$$

177 with weights normalized when their sum would otherwise exceed unity.

178 Navigation contributes through border stress:

$$\mathbf{m}_{\text{final}} = (1 - b) \mathbf{m}_{\text{primary}} + b \mathbf{m}^{N+2}. \quad (14)$$

179 N+4 therefore does not replace the existing controller. It modifies the temporal
180 conditions under which the existing cortical areas compete for behavioral influence.

181 Operational definition of the zero-persistence control

182 Equation 8 is defined only for

$$\tau_{N4} > 0.$$

183 Consequently, the condition labelled $\tau_{N4} = 0$ in the ablation experiment is an *opera-*

184 *tional zero-persistence control*, not a numerical evaluation of Eq. 8 with a zero denominator.

185 For this control, N+4 is inactive:

$$A(t) \equiv 0. \quad (15)$$

186 This corresponds to the baseline architecture in which no persistent N+4 threat state
187 contributes to behavior.

188 Human–LLM co-design protocol

189 The N+4 hypothesis emerged through a constrained dialogue between human researchers
190 and an LLM.

191 The process contained three stages.

192 *First, constraint setting.* The human authors specified the existing cortical architecture,
193 the available N+4 resource, the ecological problem, and the admissible design space.
194 Solutions involving additional sensors, replacement controllers, external memory systems,
195 or changes to the environmental rules were excluded.

196 *Second, failure analysis and hypothesis generation.* The LLM was asked what compu-
197 tation the available cortical area should perform to improve long-horizon survival. The
198 dialogue identified a temporal discontinuity: current threat information disappeared more
199 rapidly than its behavioral relevance. A persistent threat state with asymmetric rise and
200 decay was proposed as a candidate solution.

201 *Third, implementation and validation.* The human authors translated the proposed
202 function into the CM architecture, implemented the simulation, executed the experiments,
203 and performed the quantitative analyses.

204 The claim tested here is therefore not that the LLM autonomously discovered a
205 previously unknown biological mechanism. LLMs are trained on scientific text and
206 may already encode relevant concepts. The experimentally testable claim is narrower:
207 *under architectural constraints, human–LLM reasoning identified a functionally sufficient*
208 *computation for an otherwise unassigned computational resource.*

209 Evaluation and statistical analysis

210 The primary behavioral outcome is nest completion.

211 The baseline comparison uses 300 independent trials with N+4 inactive and 300
212 independent trials with N+4 active at

$$\tau_{N4} = 16.$$

213 Completion proportions are compared using Fisher’s exact test. Wilson score intervals
214 are reported for the completion proportions.

Capture counts are treated as a continuous secondary measure and compared using Welch’s unequal-variance t -test.

To examine dependence on temporal persistence, τ_{N4} is varied from the operational zero-persistence control to 32 frames:

$$0, 4, 8, 12, 16, 20, 24, 28, 32.$$

Thirty independent trials are evaluated at each condition.

A logistic curve is fitted descriptively to the observed completion proportions:

$$P_{\text{complete}}(\tau) = \frac{1}{1 + \exp[-k(\tau - \tau_c)]}, \quad (16)$$

where τ_c is the estimated midpoint of the transition.

Because the analysis is based on a finite simulation sample, the fitted transition should be interpreted as a behavioral transition in this model rather than as evidence for a formal physical phase transition.

3 Results

N+4 transforms long-horizon task completion

With N+4 inactive,

$$0/300$$

trials completed the nest, corresponding to a completion rate of

$$0\%.$$

The baseline condition also exhibited repeated capture events, with mean captures

$$8.3 \pm 2.7.$$

With N+4 active at

$$\tau_{N4} = 16,$$

nest completion increased to

$$300/300 = 100\%.$$

Mean completion time was

$$4,127 \pm 612$$

233 frames, while mean captures decreased to

$$0.4 \pm 0.7.$$

234 The difference in completion probability was significant under Fisher’s exact test,

$$p = 1.69 \times 10^{-17}.$$

235 The 95% Wilson score intervals were approximately

$$[0.00, 0.11]$$

236 for the baseline condition and

$$[0.89, 1.00]$$

237 for N+4.

238 Capture counts were also substantially reduced,

$$t_{\text{Welch}} \approx 15.5, \quad p \approx 1.1 \times 10^{-16}.$$

239 The central behavioral result is therefore not that the architecture acquired a new
 240 nest-building policy. Area N+3 already supplied that behavior. Instead, N+4 altered
 241 the temporal conditions under which the existing long-horizon behavior could remain
 242 effective.

243 **Dependence on the hysteresis timescale**

244 The completion rate depends strongly on the persistence constant τ_{N4} .

245 Table 2 reports the observed completion rates.

246 The observed curve is sigmoidal rather than linear. The descriptive logistic fit yields
 247 an estimated midpoint of approximately

$$\tau_c \approx 7.1$$

248 frames, with

$$R^2 = 0.997.$$

249 Completion approaches saturation above approximately 12–16 frames.

Table 2: Ablation over the N+4 hysteresis time constant. Thirty trials were evaluated per condition. The $\tau_{N4} = 0$ entry denotes the operational N+4-off control defined in Section 2.

τ_{N4} (frames)	Nest completion (%)
0	0
4	23
8	60
12	87
16	100
20	100
24	100
28	100
32	100

The result indicates that persistence itself is not merely an optional quantitative enhancement. Across the tested parameter range, increasing the lifetime of the threat state produces a relatively narrow behavioral transition between a regime in which the long-horizon task generally fails and one in which it is reliably completed.

Capacity and behavioral exploitability are distinct

The N+3 construction area exists in both experimental conditions. Its computational role is not created by N+4.

The difference is whether the agent can maintain behavior in the presence of temporally intermittent danger long enough for the existing construction mechanism to complete its task.

This motivates the distinction

$$\boxed{\text{computational capacity} \neq \text{behavioral exploitability}} \quad (17)$$

within the present model.

The addition of a persistent threat state does not make the construction area intrinsically more capable. It changes the temporal context in which its existing capability operates.

4 Discussion

Architectural awakening rather than within-run learning

The principal conceptual result of this work is that a major behavioral change can be produced without modifying the synaptic-like parameters of the existing controller during

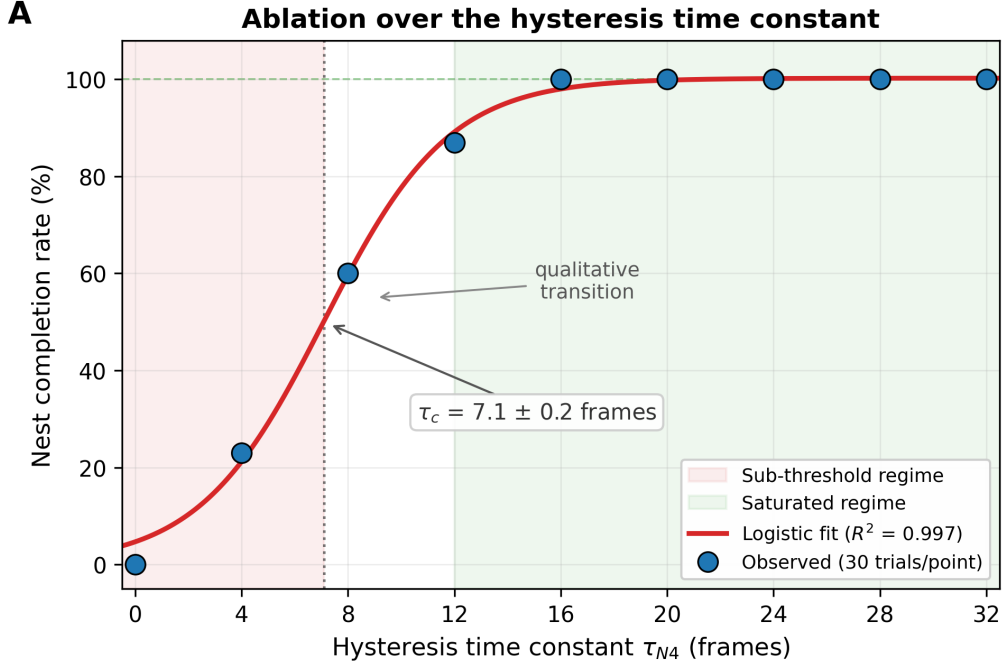


Figure 3: Ablation over the hysteresis time constant. Nest completion rate as a function of τ_{N4} , with 300 trials per condition. The observed rates follow a sharp sigmoidal transition. A descriptive logistic fit gives an estimated midpoint of approximately $\tau_c = 7.1$ frames and $R^2 = 0.997$.

269 a run.

270 The functional transformation follows the sequence

available cortical resource $\rightarrow$ functional assignment $\rightarrow$ new dynamical state $\rightarrow$ new behavioral regime.

271 This differs from a conventional learning experiment. No reward gradient trains N+4.
 272 No error signal tunes its weights. No reinforcement-learning policy is optimized from
 273 repeated failures. The mechanism is introduced through architectural allocation and then
 274 tested behaviorally.

275 The result should not be interpreted as evidence that learning or plasticity are un-
 276 necessary for intelligence. Biological intelligence clearly depends on learning at multiple
 277 scales. The narrower result is that *some* behavioral limitations may arise not because
 278 existing representations require additional training, but because a computational function
 279 required by the environment has not yet been allocated within the architecture.

280 In the present case, that function is temporal persistence.

The bottleneck is temporal in this architecture

The experiment supports the hypothesis that the relevant bottleneck in the tested environment is temporal rather than representational.

The agent already possesses a current-threat representation and a construction mechanism. Failure occurs because the influence of danger disappears too quickly relative to the environmental timescale on which the threat remains behaviorally relevant.

N+4 changes the controller from one dominated by instantaneous variables,

$$B(t) = f(X(t)), \quad (18)$$

to one in which current behavior also depends on a persistent internal state,

$$B(t) = f(X(t), A(t)), \quad (19)$$

where

$$A(t)$$

summarizes recent threat history.

This interpretation is consistent with neural accounts that emphasize persistent activity as a mechanism for maintaining information across temporal gaps (Curtis & Sprague, 2021). It is also compatible at a broader level with embodied and active-inference perspectives in which action depends on temporally extended internal dynamics rather than passive representation alone (Pezzulo et al., 2024).

The present experiment does not establish that temporal persistence is the universal bottleneck for long-horizon autonomy. It establishes that it is sufficient to resolve the specific failure observed in this architecture and environment.

Functional parallel with biological threat-memory systems

The asymmetric dynamics of N+4 exhibit a functional parallel with biological systems in which a threat continues to influence behavior after its immediate sensory trigger has changed.

Amydala-centered fear circuits support the acquisition, expression, persistence, and extinction of threat-related associations (LeDoux, 2000; Pape & Paré, 2010). Interactions between the amygdala and medial prefrontal cortex are also important in the regulation, retrieval, and extinction of fear-related states (Marek et al., 2013).

The analogy should be interpreted carefully.

N+4 is *not* claimed to be an anatomical amygdala. The 25-CM computational area does not reproduce the cellular organization of the amygdala, nor does Eq. 8 claim to

model the biochemical mechanisms underlying fear conditioning.

The relevant correspondence is computational:

- rapid response to currently relevant threat;
- persistence after immediate evidence decreases;
- modulation of competing behavior;
- a characteristic decay timescale.

The biologically interesting point is therefore one of functional convergence. A survival problem posed within the artificial architecture leads naturally to a computational requirement—persistence of recent danger—that is also important in biological threat-processing systems.

The Progressive Awakening Hypothesis

The N+4 experiment motivates a broader conjecture.

Suppose an embodied architecture contains an available computational area and the currently awakened architecture encounters a failure that cannot be resolved by its existing variables. Instead of enlarging every preceding area, adding arbitrary external memory, or retraining the entire system, the available resource can be assigned a computation that represents the missing environmental regularity.

The sequence becomes

$$N + n \rightarrow \text{new challenge} \rightarrow \text{failure} \rightarrow \text{functional hypothesis} \rightarrow N + n + 1. \quad (20)$$

We call this the *Progressive Awakening Hypothesis*:

Progressively available cortical areas may enable new computational capacities when each area is assigned a function that resolves a behavioral limitation exposed by the previously active architecture.

N+4 provides one proof of concept for this design strategy. It does not establish what N+5 or later areas should compute.

The hypothesis becomes scientifically useful only if subsequent areas are treated as falsifiable experiments. A new area should be introduced only after a new failure has been characterized, and its proposed function should be evaluated through ablation, controls, and comparison with simpler alternatives.

Human-LLM co-design as constrained hypothesis generation

The role of the LLM is also narrower than unrestricted autonomous design.

The human authors define the system, the ecological problem, the allowable resources, and the scientific constraints. The LLM operates inside this restricted design space and proposes a candidate function. The simulation then provides an external test of whether the proposed computation is useful.

The methodological loop is therefore

$$\text{architecture} \rightarrow \text{failure} \rightarrow \text{constraint} \rightarrow \text{human-LLM hypothesis} \rightarrow \text{implementation} \rightarrow \text{behavioral test} \quad (21)$$

This framing avoids the stronger and difficult-to-establish claim that the LLM independently “discovered” a biological mechanism that was absent from its training data.

Instead, the LLM functions as a constrained hypothesis-generation component in a human-controlled scientific workflow.

5 Limitations

Several limitations constrain the interpretation of the present results.

First, the simulated cortical areas are computational abstractions. Although the architecture is motivated by cortical modularity, competition, and population coding, the model should not be interpreted as a detailed reconstruction of mammalian cortical tissue.

Second, the current implementation isolates persistence of *threat magnitude* more cleanly than persistence of a complete hidden predator state. The alert variable $A(t)$ is explicitly gated by the threat-perception signal, whereas directional information used by the escape vector is represented separately. A stronger future test should freeze the last perceived predator bearing when $\pi(t)$ falls below the perceptual range and require N+4 to operate only from that remembered bearing. This would separate persistent threat memory from continued access to directional world-state information.

Third, the experiment demonstrates functional sufficiency but not uniqueness or necessity. A scalar hysteretic variable outside a 25-CM area might also improve performance. A direct control comparing the complete N+4 implementation with a scalar-only temporal buffer is therefore required to determine what is specifically contributed by the cortical-area implementation.

Fourth, the ecological setting is deliberately simple. The result has been demonstrated for intermittent predation combined with a nest-construction task. Generalization should be tested under different predator statistics, temporary occlusion, intermittent hazards, delayed consequences, and tasks whose relevant temporal scale differs from the present

environment.

Fifth, the ablation uses 30 trials per condition. The observed effect is large, but larger preregistered simulation sets, archived random seeds, and trial-level data would provide a more rigorous estimate of robustness.

Finally, the LLMs involved in the co-design process were trained on scientific and technical text and may have encoded prior knowledge of hysteresis, working memory, control systems, or fear circuitry. The experiment therefore does not establish independent biological discovery by an LLM. Its methodological contribution is the use of constrained human-LLM reasoning to transform prior knowledge into a testable functional assignment within a fixed architecture.

6 Future Experiments

The present model generates several testable extensions.

A first experiment should vary the characteristic interval between threat disappearance and reappearance. If N+4 is genuinely adapting behavior to an environmental temporal regularity, the optimal persistence parameter should covary with that interval:

$$\tau_{N4}^* \sim T_{\text{threat}}, \quad (22)$$

where T_{threat} is the timescale over which previous predator presence remains behaviorally informative.

A second experiment should remove access to current predator direction outside the perceptual radius and retain only the last observed bearing. If the benefit of N+4 remains, the interpretation as a genuine temporal bridge across perceptual absence becomes substantially stronger.

A third experiment should compare four conditions: reactive control, scalar threat hysteresis, 25-CM N+4 hysteresis, and a generic non-threat-specific memory variable. This would distinguish the contributions of temporal persistence, threat specificity, and cortical population structure.

Finally, the Progressive Awakening Hypothesis should be tested prospectively. Rather than assigning N+5 a desired high-level capability in advance, a new ecological challenge should first be introduced. Only after the N+4 architecture exhibits a reproducible failure should human-LLM co-design be used to propose a computational function for the next available area.

7 Conclusion

This study began with an embodied-agent failure rather than with a new learning algorithm.

The baseline architecture already possessed reactive control, navigation, and nest-construction capability, yet intermittent danger prevented reliable completion of the long-horizon task.

A constrained human-LLM co-design process identified a missing computation: temporal persistence of recent threat.

Assigning this computation to an available cortical area, N+4, produced a sharp behavioral transformation. Under the reported experiment, nest completion increased from 0% to 100%

while capture events were strongly reduced.

The N+4 mechanism was not acquired through gradient descent or within-run synaptic adaptation. It was introduced through functional allocation of an available architectural resource.

The central result can therefore be summarized as

$$\boxed{\text{available computational resource} + \text{appropriate functional allocation} \rightarrow \text{new behavioral regime}} \quad (23)$$

This does not imply that learning is unnecessary. It identifies a complementary route to behavioral complexity: some capabilities may emerge when previously inactive computational resources are assigned functions that represent environmental regularities missing from the preceding architecture.

N+4 provides one experimentally testable instance of this idea.

The broader Progressive Awakening Hypothesis now asks whether subsequent behavioral failures can similarly reveal what computations become necessary for N+5, N+6, and later available areas.

Data availability statement

The simulator code and current study materials are openly available in the *Cortical-Microcircuits-Design* repository at:

<https://github.com/ochantor/Cortical-Microcircuits-Design>

Trial-level results, random seeds, and the batch/statistical-analysis scripts corresponding to Table 2 and Figure 3 will be deposited in the same repository as part of the versioned submission release.

AI-use declaration

Large language models, including Claude 3.5 Sonnet (Anthropic) and GPT-4 (OpenAI), were used as components of the conceptual co-design process and for language-assisted

manuscript development. The human authors defined the architecture and constraints, selected and interpreted the functional hypotheses, implemented the simulation code, executed the experiments, performed the statistical analyses, and take responsibility for all scientific claims and conclusions. LLM output was treated as hypothesis-generating material rather than as experimental evidence.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The authors acknowledge the use of large language models during the constrained human–LLM co-design process described in this work.

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